Images I_t

Text

Lift the long thin bottle using the right arm and hold upright

3D Marker Token

VLM Backbone

Obs Embedding

Geo Embedding

Event Guide Memory

Scene Change & Event Novelty

Adaptive Gate

Memory Bank

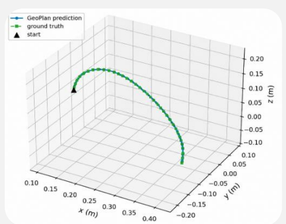
Train only

Noise

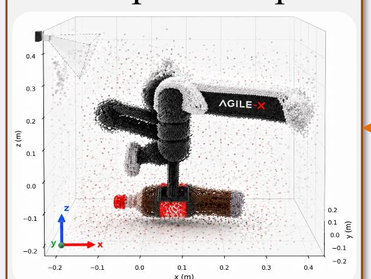
3D-Spatial Plan Head

VGGT

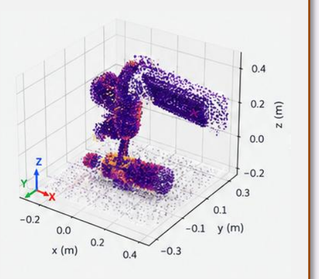
EEF Tracks



3D point-map



3D Scene GT



EEF GT

Memory Fusion

Noise

Action Expert



VLM Modules

Memory Modules

3D Plan Modules